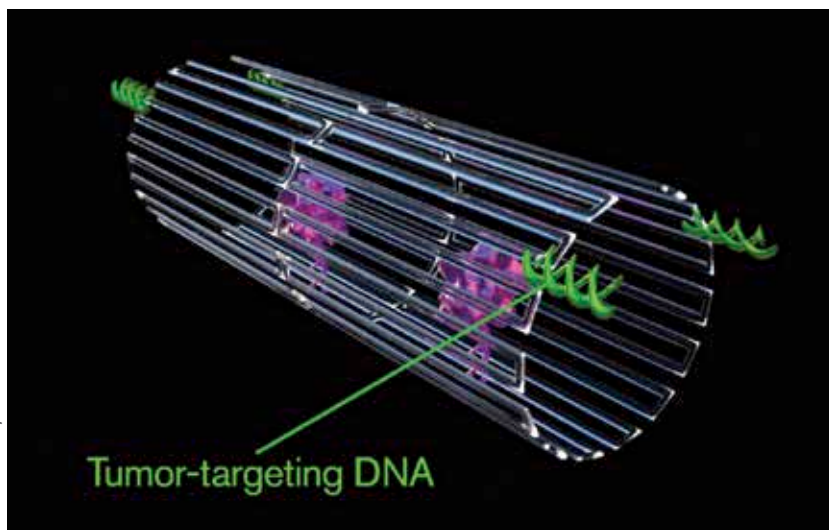


Credit: University of Arizona



An example of a tumour fighting nanobot

For starters, diagnostics will be a whole new ballgame and will involve far less guesswork than now. We like to think our blood tests and imaging techniques are remarkable, and perhaps they are, but only compared to the past. The future will do away with harmful imaging such as x-rays and CT scans, and instead our bodies will be diagnosed from the inside out. Feeling sickly for some time now? All you will need to do is swallow a capsule filled with nanosensors that will spread throughout your body and create the most comprehensive report about you that has ever been done. And it won't be guesswork based on blood reports either... you will have proper data from inside your actual kidneys, telling you how they're

working. Nano-capsules inside your stomach will test the gases and pH levels, those in your bloodstream will monitor for blockages and irregularities, and diagnostics will essentially be handed over to a doctor's computer.

Drug delivery will also be done through nanoscale bots, which will carry tiny amounts of medicines and target delivery to the specifically inflamed or afflicted areas of a body, instead of the current spray and pray method of drug delivery. This is especially good for antibiotics, as we might be able to target them at a particular infection zone, rather than expose the whole body to the antibiotic and risk bacteria forming an immune response via mutation. It's not just delivery, but also monitoring

of doses and ensuring that the right amount of drugs are being ingested.

Then of course there is the promise of nanobots, which will not be miniature versions of terminator looking robots. It's far more likely that we will follow nature's lead on this and develop a T-cell bot, for instance, which will attack cancer cells. While cancer cells are able to 'hide' from our own bodies defenses, they will not be able to do so from nanobots that have been programmed to find them, and have a human doctor guiding them. The 'cure' for cancer might just be a billion n-cell (we're being super original by replacing the T in t-cells with n for nano).

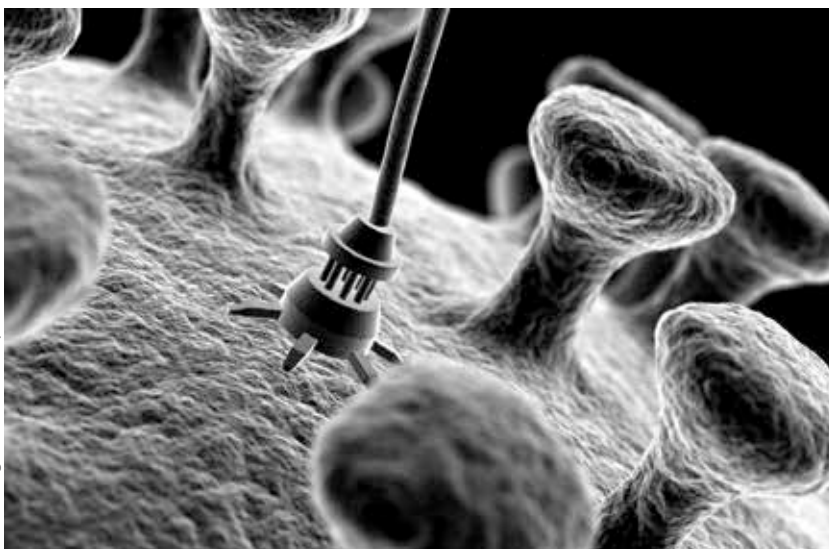
We may also finally find a cure for viruses using nanotechnology, using the same n-cells that can attack cancer growths. We'd just need to program those nanocells to find and attack HIV, or COVID-19, or just about any viral infection contained in a body.

Those suffering from blockages in arteries will also no longer need surgery or other invasive methods to break down arterial plaque. Instead, a simple injection into a vein will do the trick. You sit in an office watching TV or reading while a doctor analyses your blood system and then directs nanobots to start attacking the plaque buildup in an artery much like piranhas attack a piece of meat. Thousands of these little bots will all 'bite off' a nano chunk and swim away, awaiting their turn to attack the plaque monster again.

Space exploration

Our biggest problem with space exploration is actually getting there. Overcoming earth's gravity is the biggest hurdle we have. If it wasn't for the horrible cost of sending stuff into space, we'd have happy go-lucky chaps driving off to visit Mars in homemade spacecraft on a regular basis. People are already jumping out of aeroplanes and off cliffs just to get an adrenaline rush, can you imagine what space would be like if just anyone could go? You may not need to imagine that in the distant future, but for now, the challenge is to be able to get tonnes of equipment and fuel up into space,

Credit: George Mason University



A nano probe that can grasp, constrain and move individual cells